

MSB7216: Deep Learning for Health Data

Final Project Report

Deep Learning for Automated Sickle Cell Detection in Peripheral Blood Smears:

An Explainable Approach Using African Clinical Microscopy Data

Okidi Patrovas Gaabriel

Student ID: 2025/HD07/26020U

Makerere University, Kampala, Uganda

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Live Demo: <https://huggingface.co/spaces/Patrovas/sickle-cell-detection>

GitHub: <https://github.com/Patro331/sickle-cell-detection>

1. Abstract

Sickle cell disease is one of the most prevalent genetic disorders in Uganda, with the sickle cell trait estimated at 13.3% nationally and reaching 19.8% in Kampala. Diagnosis depends on trained laboratory personnel and specialist microscopy equipment that are largely unavailable in rural settings. This project presents the first deep learning study to apply transfer learning and explainability to the (Tushabe et al., 2024) Ugandan clinical mobile-phone microscopy dataset for automated sickle cell detection. A pipeline of three models was developed and evaluated: a baseline Convolutional Neural Network trained from scratch, EfficientNet-B0 with transfer learning, and ResNet-50 with transfer learning. Using a combined dataset of 933 blood smear images, ResNet-50 achieved the best performance with 98.57% accuracy, AUC-ROC of 0.9973, sensitivity of 98.41%, and specificity of 98.70% on a held-out test set of 140 images. Grad-CAM explainability analysis confirmed the model attends to biologically meaningful regions. The final model was deployed as a Gradio application on Hugging Face Spaces, demonstrating practical viability as a low-resource screening tool for rural Ugandan health workers.

Keywords: *Sickle Cell Disease, Deep Learning, Transfer Learning, ResNet-50, EfficientNet-B0, Grad-CAM, Medical Image Classification, Uganda, Low-Resource Healthcare*

2. Introduction

Sickle cell disease is a monogenic blood disorder caused by a mutation in the HBB gene, resulting in the production of haemoglobin S. Under deoxygenated conditions, haemoglobin S polymerises and distorts red blood cells into a characteristic crescent or sickle shape. This morphological change causes vaso-occlusion, haemolytic anaemia, and progressive organ damage. Uganda carries a disproportionately high burden of sickle cell disease, with an estimated 13.3% of the population carrying the sickle cell trait and prevalence reaching 19.8% in Kampala alone. An estimated 15,000 to 20,000 children are born with the disease in Uganda each year, many of whom go undiagnosed until a severe crisis occurs.

Current diagnosis relies on laboratory-based methods including peripheral blood smear microscopy, haemoglobin electrophoresis, and the sickling test. These approaches require trained laboratory scientists, specialist equipment, and reliable staining supplies that are largely unavailable in rural district hospitals and community health centres. The result is a significant diagnostic gap: patients in rural Uganda may go years without a formal diagnosis, missing the early interventions such as prophylactic penicillin and hydroxyurea therapy that substantially reduce morbidity and mortality.

Deep learning has demonstrated strong performance in medical image classification tasks including diabetic retinopathy screening, tuberculosis detection from chest X-rays, and malaria parasite identification from blood smears. These technologies offer the possibility of deploying specialist-level diagnostic capability on low-cost mobile devices, bridging the resource gap in low-income settings. However, the vast majority of deep learning medical imaging studies use datasets collected under controlled laboratory conditions in high-income countries, limiting their applicability to the imaging contexts of sub-Saharan Africa.

This project addresses this gap directly. We present the first deep learning study to apply explainable transfer learning to the (Tushabe et al., 2024) Ugandan Sickle Cell Microscopy Dataset, a collection of blood smear images captured using mobile phone cameras placed on basic microscopes in Soroti and Kumi districts, Uganda. The primary contributions of this work are as follows.

- Development of a complete deep learning pipeline for binary sickle cell classification using clinically realistic Ugandan microscopy images.
- Systematic comparison of three architectures: a baseline CNN trained from scratch, EfficientNet-B0 with transfer learning, and ResNet-50 with transfer learning.
- Grad-CAM explainability analysis across all four prediction categories, including detailed error analysis of both misclassified test cases.
- Deployment of the best model as a Gradio application on Hugging Face Spaces, demonstrating practical viability as a mobile-first screening tool.
- Critical discussion of dataset limitations, domain gaps, and ethical considerations relevant to deployment in low-resource Ugandan settings.

3. Related Work

3.1 Deep Learning for Medical Image Classification

Deep learning and convolutional neural networks have transformed medical image analysis over the past decade (Esteva et al., 2019). demonstrated dermatologist-level skin cancer

classification using a deep convolutional network, establishing that deep learning could match specialist human performance in diagnostic imaging tasks. (Rajpurkar et al., 2017) showed that DenseNet-121 could detect pneumonia from chest X-rays with radiologist-level accuracy. These foundational studies established the viability of deep learning as a clinical decision support tool.

3.2 Transfer Learning in Medical Imaging

Training deep learning models from scratch requires large labelled datasets that are rarely available in medical imaging contexts. Transfer learning addresses this limitation by leveraging feature representations learned from large general-purpose datasets such as ImageNet and fine-tuning them for specific medical tasks. (Tajbakhsh et al., 2016) demonstrated that fine-tuned pretrained networks consistently outperformed networks trained from scratch across four medical imaging tasks, even with limited training data. (Matsoukas, Haslum, Sorkhei, Söderberg, & Smith, 2022) specifically examined transfer learning for medical imaging and found that features from ImageNet-pretrained networks transferred effectively to medical tasks despite the visual domain difference.

3.3 Deep Learning for Blood Cell Analysis

Automated blood cell analysis using deep learning has been an active research area since the availability of high-quality digital microscopy datasets. (Acevedo et al., 2020) published a dataset of 17,092 annotated peripheral blood cell images and demonstrated automated classification across eight cell types. (Asghar, Kumar, Hynds, & Shaukat, 2023) applied deep learning to white blood cell classification with over 98% accuracy. More recently, attention-based architectures have been applied to blood cell morphology analysis, demonstrating the richness of learnable features in microscopy images.

3.4 Sickle Cell Detection with AI

Several studies have specifically addressed automated sickle cell detection. (Qiang, Xu, Pochron, Jupelli, & Dao, 2024) applied a convolutional network to sickle cell detection in microfluidic deformability assay images. (Poostchi et al., 2018) reviewed machine learning approaches for malaria parasite detection in blood smear images, establishing the feasibility of mobile-phone microscopy as an imaging platform for automated analysis. However, all prior sickle cell deep learning studies used datasets collected under controlled laboratory conditions in high-income countries. None used mobile-phone microscopy data from clinical settings in sub-Saharan Africa.

3.5 AI in Low-Resource Healthcare Settings

The application of deep learning to healthcare in low-resource settings has received increasing attention. (Arowosegbe & Oyelade, 2022) reviewed AI applications in sub-Saharan African healthcare, identifying a critical gap between the geographic distribution of AI model development and the regions with the greatest diagnostic burden. (Abayomi-Alli, Misra, & Abayomi-Alli, 2022) highlighted the need for datasets collected under real-world conditions in African healthcare settings, noting that models trained on images from high-income country hospitals show degraded performance when deployed in low-resource contexts. This project directly addresses this recommendation by using a dataset collected from Ugandan patients using the same mobile-phone microscopy setup that would be used in real-world deployment.

3.6 Explainability in Medical AI

The adoption of deep learning in clinical settings has been limited in part by the difficulty of interpreting neural network predictions. (Selvaraju et al., 2017) introduced Gradient-weighted Class Activation Mapping (Grad-CAM), which produces visual explanations by computing the gradient of the predicted class score with respect to the final convolutional layer feature maps. Grad-CAM has been widely applied in medical imaging to validate that models attend to clinically relevant regions, increasing clinician trust and facilitating responsible deployment. This project applies Grad-CAM to provide visual explanations for all predictions, a critical component for responsible deployment in clinical settings.

4. Dataset Description

4.1 Initial Dataset Selection and Revisions

The dataset selection process went through two revisions before arriving at the final combination used for training. Understanding these revisions is important because they reflect real-world challenges in assembling medical imaging datasets for low-resource settings.

The first planned combination was the (Tushabe et al., 2024) sickle cell dataset paired with the Mendeley Peripheral Blood Cell Images dataset by (Acevedo et al., 2020) as the normal class. However, upon detailed inspection of the Mendeley dataset, it became clear that it contained segmented images of white blood cells and platelets, specifically lymphocytes, neutrophils, eosinophils, monocytes, basophils, erythroblasts, and platelets, rather than normal red blood cell smear images. Using white blood cells as the negative class in a sickle cell detection task would have introduced a fundamental biological inconsistency into the training data: the model would have been trained to distinguish sickle cells from the wrong cell type entirely. The Mendeley dataset was therefore replaced with the BCCD Dataset, which contains actual blood smear photographs showing normal red blood cells in their natural context.

4.2 Dataset One: Tushabe et al. (2024) Uganda Sickle Cell Dataset

The primary dataset is the (Tushabe et al., 2024) Ugandan Sickle Cell Microscopy Dataset, published in *Acta Scientific Microbiology*, Volume 7, Issue 12. It contains blood smear images collected from patients in Soroti and Kumi districts, Uganda, using mobile phone cameras placed on the eyepiece of basic optical microscopes. This imaging approach is representative of the equipment available in low-resource Ugandan clinical settings.

The dataset was originally divided into two subsets: a labelled subset and an unlabelled subset. The labelled subset contained bounding box annotations marking the locations of individual sickle cells within each image. However, a critical data quality issue was discovered during visual inspection of augmented training samples: the bounding box annotations in the labelled subset were burned directly into the image pixels themselves, rather than being stored as separate annotation files. This meant that the bounding box outlines appeared as visible black rectangles drawn on top of the blood smear images. If these images were used for classification training, the model would learn to associate the presence of black rectangular boxes with the positive sickle cell class, rather than learning the actual crescent-shaped morphology of sickle cells. This would produce a model that performs well on the annotated

training data but fails completely in real-world deployment where no bounding boxes are drawn on images.

To prevent this, all 422 labelled images were excluded from training. Only the 422 unlabelled positive images, which contained no burned-in annotations, were retained for the positive class. This decision reduced the available positive training data from 844 images to 422 images but was necessary to ensure the model learned genuine biological features.

The dataset also includes 147 normal blood smear images in the negative subset. These images were captured under the same mobile-phone microscope conditions as the positive images and therefore share the same characteristic circular border produced by the phone camera being placed over the microscope eyepiece. These 147 images were retained as part of the negative class.

4.3 Dataset Two: BCCD Dataset

The supplementary dataset is the BCCD (Blood Cell Count and Detection) Dataset by Shenggan et al., distributed under the MIT License on GitHub. It contains 364 normal peripheral blood smear photographs captured under controlled laboratory conditions. These images served as the negative class supplement, providing additional examples of normal blood smears for the classifier. Unlike the Tushabe images, BCCD images were taken with a standard laboratory camera rather than a mobile phone and therefore do not have the circular border characteristic of the Tushabe images. This difference in appearance between the two negative class data sources represents a known domain difference within the negative class that is discussed further in the Limitations section.

4.4 Final Dataset Composition

The final combined dataset consisted of 933 images across the two classes. The positive class contained 422 unlabelled sickle cell images from the Tushabe dataset. The negative class contained 511 images in total, comprising 147 normal images from the Tushabe dataset and 364 images from the BCCD dataset.

Source	Class	Images	Camera Type	Annotation Status
Tushabe (2024) unlabelled	Positive	422	Mobile phone	No annotations
Tushabe (2024) labelled	Positive	422 excluded	Mobile phone	Burned-in boxes
Tushabe (2024)	Negative	147	Mobile phone	Clean
BCCD Dataset	Negative	364	Lab camera	Clean
Total used	Both	933	Mixed	Clean

Table 1: Dataset composition showing all sources, the exclusion of labelled images, and final sample counts.

4.5 Dataset Split

The dataset was split into training, validation, and test sets using a stratified 70/15/15 ratio. Stratification ensured that the class proportions were preserved across all three sets. The final split resulted in 653 training images, 140 validation images, and 140 test images.

5. Methodology

5.1 Exploratory Data Analysis

Comprehensive exploratory data analysis was conducted prior to model development. Key findings included significant image size variation, with positive images ranging from 990x750 to 3122x3094 pixels and negative images ranging from 640x480 to 1000x1116 pixels, reflecting the different imaging conditions between the two data sources. Pixel intensity analysis revealed comparable mean intensities between classes (159.6 for positive, 157.0 for negative), suggesting that brightness alone would not be a strong discriminating feature. All 933 images passed a quality check with zero corrupted, non-RGB, or undersized files identified.

5.2 Preprocessing Pipeline

All images were resized to 224 by 224 pixels to match the input requirements of EfficientNet-B0 and ResNet-50. Normalisation was applied using ImageNet mean values of 0.485, 0.456, and 0.406 and standard deviation values of 0.229, 0.224, and 0.225, as both pretrained architectures were originally trained on ImageNet.

Data augmentation was applied exclusively to the training set using a clinically motivated strategy. Random horizontal and vertical flips were applied since blood smear cells appear at any orientation. Random rotation up to 180 degrees addressed the absence of a fixed microscope slide orientation. Colour jitter with brightness, contrast, saturation, and hue variation simulated the lighting inconsistency characteristic of mobile phone cameras under field conditions. Random resized cropping simulated variation in microscope zoom level.

Class imbalance was addressed through two complementary strategies. A WeightedRandomSampler was applied to the training DataLoader, assigning each class a weight inversely proportional to its count. A weighted Binary Cross-Entropy loss function with pos_weight of 1.2061 was used during training to further penalise misclassification of the minority class.

5.3 Model Architectures

Three model architectures were developed and compared.

The baseline model was a simple three-block convolutional network trained entirely from scratch with no pretrained weights. Each block comprised a convolutional layer, batch normalisation, ReLU activation, and max pooling, with filter counts of 32, 64, and 128 respectively. The classifier head used dropout regularisation followed by two fully connected layers. This model contained 25,784,321 trainable parameters and served as a performance floor.

EfficientNet-B0 pretrained on ImageNet was selected for its compound scaling design, which uniformly scales network depth, width, and resolution to achieve strong performance with relatively few parameters, totalling 4,008,829. The original classifier head was replaced with a binary output layer with 0.3 dropout. Both frozen backbone and unfrozen backbone strategies were evaluated, with the unfrozen approach selected as the primary model based on validation performance.

ResNet-50 pretrained on ImageNet was selected for its residual connections, which allow gradients to flow through deep networks and have proven effective in medical image classification tasks. The architecture contains 23,510,081 parameters. The final fully connected layer was replaced with a binary classifier with 0.3 dropout. Both frozen and unfrozen backbone strategies were evaluated, with the unfrozen approach selected.

5.4 Training Configuration

All models were trained using the Adam optimiser with a learning rate of 0.001 for baseline and frozen transfer learning models, and 0.0001 for unfrozen transfer learning models. The lower learning rate for fine-tuning is standard practice to avoid overwriting pretrained representations through large gradient updates. A ReduceLROnPlateau scheduler reduced the learning rate by a factor of 0.5 when validation loss did not improve for five consecutive epochs. Early stopping with a patience of 10 epochs was applied throughout, saving the best model weights based on minimum validation loss. All experiments were conducted on a Tesla T4 GPU via Google Colab.

5.5 Grad-CAM Explainability

Gradient-weighted Class Activation Mapping was applied to ResNet-50 to visualise the regions of each input image that most influenced the model's prediction. PyTorch backward hooks were registered on the final convolutional block, referred to as layer4, to capture gradients and feature map activations. The Grad-CAM heatmap was computed by weighting each feature map channel by the mean gradient of the predicted class score with respect to that channel, applying ReLU to retain only positive contributions, and resizing to the input image dimensions. Heatmaps were generated for examples from all four prediction categories: true positives, true negatives, false positives, and false negatives.

6. Experiments

6.1 Experiment Design

Three experiments were conducted in sequence. Each experiment used the same training, validation, and test sets to ensure fair comparison. All random seeds were set to 42 throughout to ensure reproducibility. Training and validation loss and accuracy were recorded at each epoch to enable analysis of learning behaviour and overfitting.

For the two transfer learning architectures, an ablation study was conducted comparing frozen backbone and unfrozen backbone strategies. In the frozen strategy, all pretrained convolutional layers were fixed and only the new classifier head was updated during training. In the unfrozen strategy, all layers including the pretrained convolutional layers were updated during training. This comparison allowed us to quantify the benefit of adapting pretrained features to the blood smear microscopy domain.

6.2 Hyperparameter Justification

Hyperparameter	Value	Justification
Learning rate (baseline and frozen)	0.001	Standard for training from scratch

Learning rate (unfrozen fine-tuning)	0.0001	Smaller updates preserve pretrained features
Batch size	32	Balance between memory and gradient stability
Maximum epochs	50	Sufficient with early stopping patience of 10
Optimiser	Adam	Adaptive learning rate suits medical imaging tasks
Loss function	Weighted BCEWithLogitsLoss	Addresses class imbalance in the training set
Input image size	224 x 224 pixels	Standard input size for EfficientNet and ResNet
Dropout rate	0.3 on classifier head	Regularisation to reduce overfitting on small dataset

Table 2: Hyperparameters used across all experiments with justifications.

6.3 Evaluation Metrics

Models were evaluated on the held-out test set of 140 images using five primary metrics. Accuracy measures the overall proportion of correctly classified images. AUC-ROC measures discriminative ability across all decision thresholds and is particularly informative for imbalanced datasets. F1 Score measures the harmonic mean of precision and recall. Sensitivity, which is the recall for the positive class, is the most clinically critical metric since missing a sickle cell patient carries greater health consequences than incorrectly referring a normal patient. Specificity measures the proportion of normal cases correctly identified. Confusion matrices were generated for all models to provide a complete breakdown of prediction errors.

6.4 Training Outcomes

The baseline CNN trained for 37 epochs before early stopping was triggered, with the best model saved at epoch 27. The frozen EfficientNet-B0 trained for 18 epochs with the best model at epoch 8. The unfrozen EfficientNet-B0 trained for 22 epochs with the best model at epoch 12. The frozen ResNet-50 trained for 41 epochs with the best model at epoch 31. The unfrozen ResNet-50 trained for 25 epochs with the best model at epoch 15.

The ablation comparison between frozen and unfrozen strategies consistently favoured the unfrozen approach. For EfficientNet-B0, the unfrozen best validation loss of 0.2112 was substantially lower than the frozen best validation loss of 0.2973. For ResNet-50, the unfrozen best validation loss of 0.2354 was lower than the frozen best validation loss of 0.2510. This confirms that adapting pretrained features to the blood smear microscopy domain through full fine-tuning produces meaningfully better representations than using pretrained features as a fixed extractor.

7. Results

7.1 Comparative Performance

Model	Accuracy	AUC-ROC	Sensitivity	Specificity	F1 Score	Errors
Baseline CNN	87.86%	0.9588	95.24%	81.82%	0.8759	17
EfficientNet-B0	96.43%	0.9951	95.24%	97.40%	0.9600	5
ResNet-50	98.57%	0.9973	98.41%	98.70%	0.9841	2

Table 3: Comparative performance of all three models on the held-out test set of 140 images.

7.2 Baseline CNN Results

The baseline CNN achieved 87.86% accuracy, AUC-ROC of 0.9588, sensitivity of 95.24%, and specificity of 81.82% on the test set. The model correctly identified 60 out of 63 sickle cell cases and 63 out of 77 normal cases. The 14 false positives reflected the model's difficulty in reliably distinguishing normal smears without pretrained feature representations. Training curves showed characteristic overfitting, with training accuracy climbing to 96.17% while validation accuracy fluctuated noisily around 87 to 90%, confirming that training from scratch on 653 training images produces unstable feature representations.

7.3 EfficientNet-B0 Results

The unfrozen EfficientNet-B0 achieved accuracy of 96.43%, AUC-ROC of 0.9951, sensitivity of 95.24%, and specificity of 97.40%. False positives dropped from 14 in the baseline to just 2, demonstrating that pretrained ImageNet features provided far more reliable representations for distinguishing normal smears. False negatives remained at 3, the same as the baseline. Training curves were notably smoother and more stable than the baseline, reflecting the advantage of pretrained feature representations.

7.4 ResNet-50 Results

The unfrozen ResNet-50 achieved the best performance across all metrics: accuracy of 98.57%, AUC-ROC of 0.9973, F1 score of 0.9841, sensitivity of 98.41%, and specificity of 98.70%. Only 2 errors were made on 140 test images, comprising 1 false positive and 1 false negative. Despite exhibiting noisier validation curves during training than EfficientNet-B0, attributable to its larger parameter count being fine-tuned on a limited dataset, ResNet-50 generalised exceptionally well to the held-out test set.

The improvement in sensitivity from 95.24% in the baseline to 98.41% in ResNet-50 is clinically significant. In a theoretical screening of 1,000 patients with 13.3% sickle cell trait prevalence, approximately 133 sickle cell cases would be expected. ResNet-50 would miss only 2 of these cases, compared to 6 missed by the baseline CNN.

8. Error Analysis

8.1 Overview of Errors

ResNet-50 made 2 errors on 140 test images: 1 false positive and 1 false negative. Grad-CAM analysis was applied to both misclassified images to understand what the model attended to when making incorrect predictions.

8.2 False Positive Analysis

The single false positive was a normal blood smear image that the model predicted as sickle cell positive with a confidence of 0.841. Examination of the original image revealed a small dark particle visible near the upper region of the circular microscope field. This particle was likely a dust particle, a staining precipitate, or an optical artefact introduced during sample preparation or imaging.

The Grad-CAM heatmap showed the model's attention concentrated entirely on this artefact rather than on any cellular morphology. The model was not making a biology-based prediction: it was responding to a visual anomaly that superficially resembled the kind of localised dark feature associated with abnormal cell presentations. This type of error is a known challenge in mobile-phone microscopy, where dust, staining inconsistencies, and optical artefacts can occur more frequently than in controlled laboratory settings.

The high confidence of 0.841 on this incorrect prediction is a concern. It suggests the model did not have sufficient uncertainty to flag this case for review. One approach to mitigating this failure mode would be to implement an artefact detection preprocessing step that identifies and masks imaging anomalies before the classification model receives the image.

8.3 False Negative Analysis

The single false negative was a sickle cell positive image that the model predicted as normal with a confidence of 0.472, just below the 0.5 decision threshold. The Grad-CAM heatmap showed diffuse activation across the lower portion of the image without a clear focal point, indicating that the model was genuinely uncertain about this case rather than confidently wrong.

Visual inspection of the original image revealed a densely packed blood smear with unusual pink staining compared to the majority of positive training images. The dense cell packing made it difficult to identify individual sickle-shaped cells clearly, and the staining variation may have shifted the pixel intensity distribution away from what the model had learned to associate with positive cases.

The low confidence of 0.472 provides an actionable path forward. Lowering the decision threshold from 0.5 to 0.4 would classify this image as positive, eliminating the false negative at the cost of potentially one additional false positive. In a clinical screening context where sensitivity is more important than specificity, a threshold of 0.4 would be more appropriate than the default 0.5.

8.4 Grad-CAM Findings Across All Predictions

Across the full set of 140 test images, Grad-CAM analysis revealed consistent patterns by prediction category. True positive images showed concentrated focal activation within the circular microscope field, aligned with cell-dense regions containing visible sickle-shaped cells. True negative images showed diffuse activation with no strong focal point, consistent with the absence of distinctive pathological features. The contrast between concentrated activation in true positives and diffuse activation in true negatives confirms that the model has learned a meaningful asymmetry rooted in the biology of the task.

9. Ethical Considerations

This project raises several ethical considerations that must be addressed before any clinical deployment.

This model is explicitly a screening tool and must never be presented as a diagnostic replacement for trained clinical judgement. All positive predictions must be confirmed by a qualified medical professional before any clinical action is taken. The Gradio application includes a prominent disclaimer to this effect. Any deployment must be accompanied by clear communication to health workers about the tool's role and limitations.

Non-specialist health workers using this tool may not have sufficient background to contextualise model predictions or recognise when an output should be questioned. Training and supervision protocols would be essential in any deployment to ensure the tool augments rather than replaces clinical reasoning.

Both datasets used in this study were collected with documented patient consent and are freely available for research and educational use. The (Tushabe et al., 2024) paper documents the consent procedures followed for Ugandan patient data collection. No personally identifiable information is stored or processed by the deployment application.

The model was trained and evaluated exclusively on images from Ugandan patients in Soroti and Kumi districts. Performance on patients from other regions of Uganda, other East African countries, or other populations is unknown. Any deployment beyond the training context should be preceded by validation on locally collected data.

The explicit framing of this tool for low-resource Ugandan health workers aligns with health equity principles. However, unequal access to smartphones and basic microscopes within Uganda means the tool may be more accessible to better-resourced district hospitals than to the most isolated rural communities with the greatest diagnostic need.

10. Limitations

Several limitations of this study must be acknowledged.

The first limitation relates to the initial dataset selection. The originally planned normal class dataset, the Mendeley Peripheral Blood Cell Images by (Acevedo et al., 2020), was found upon inspection to contain segmented white blood cells and platelets rather than normal red blood cell smear images. Using this dataset would have introduced a fundamental biological inconsistency into the model since the negative class would not have represented normal blood smears at all. The dataset was replaced with the BCCD Dataset, which was a more appropriate choice. However, this replacement introduced a domain gap described further below.

The second limitation relates to the exclusion of labelled training data. The (Tushabe et al., 2024) labelled subset contained 422 images with bounding box annotations that were burned directly into the image pixels rather than stored as separate annotation files. These annotations appeared as visible black rectangles drawn on top of the blood smear images. If used for training, the model would have learned to identify black rectangular boxes as a feature of the positive class rather than learning the actual morphology of sickle cells. All 422 labelled images were therefore excluded, reducing the positive training set from 844 images to 422 images and halving the available positive training data. Future data collection efforts should store annotations as separate files rather than drawing them on the image itself.

The third limitation is the domain gap between the two negative class data sources. Tushabe images carry a characteristic circular border produced by the mobile phone camera being placed over the microscope eyepiece, while BCCD images were captured with a standard laboratory camera and appear as standard rectangular photographs with no circular border. This visual difference exists only within the negative class and therefore cannot directly confuse the positive versus negative classification. However, it represents a visual inconsistency in the training data that may affect the model's performance on negative cases from imaging contexts not represented in the training set. Future work should source additional normal smear images captured using mobile-phone microscope conditions consistent with the Tushabe dataset to create a more visually uniform negative class.

The fourth limitation is the overall dataset size of 933 raw images, which is small by deep learning standards. While augmentation expanded the effective training set and transfer learning substantially mitigated the small data problem, the results may not generalise reliably to patient populations from regions other than Soroti and Kumi districts. External validation on an independently collected Ugandan dataset would be required before clinical deployment.

The fifth limitation is that this model has not undergone clinical validation. All reported performance metrics are based on a held-out test set drawn from the same distribution as the training data. Performance in a real clinical deployment may differ due to differences in microscope models, phone cameras, staining reagents, and laboratory technician experience.

The sixth limitation is the single false positive caused by a staining artefact. This highlights a vulnerability to imaging noise that is not addressed by the current model architecture. An artefact detection preprocessing step would be required to reduce this error mode in deployment.

11. Future Work

Several directions for future work follow directly from the findings and limitations of this study.

External clinical validation is the most important next step. The model should be evaluated on an independently collected dataset of blood smear images from Ugandan clinical settings before any deployment. This validation should include images from multiple districts, different microscope models, different phone cameras, and different staining protocols to assess generalisability.

Artefact detection preprocessing would directly address the false positive failure mode identified in this study. A preprocessing module that detects and masks imaging anomalies such as dust particles, staining precipitates, and optical artefacts before the classification model receives the image would reduce the risk of artefact-driven false positives.

Decision threshold optimisation should be explored for deployment. The analysis of the false negative case showed that a threshold of 0.4 rather than 0.5 would eliminate the error at the cost of one additional false positive. In a screening context where sensitivity is the priority, the optimal threshold should be determined through analysis of the full precision-recall trade-off curve rather than defaulting to 0.5.

A more visually uniform negative class dataset would strengthen the model. Collecting or sourcing normal blood smear images captured using mobile-phone microscopes under conditions similar to the Tushabe dataset would eliminate the circular border domain gap that currently exists within the negative class.

Extension to multi-class haematological disorder classification is a natural progression. The same mobile-phone microscopy framework could be applied to classify multiple blood disorders simultaneously, increasing the clinical utility of the tool for health workers in low-resource settings.

12. Conclusion

This project developed and evaluated an explainable deep learning pipeline for automated sickle cell detection from peripheral blood smear images, with specific application to low-resource clinical settings in Uganda. Using a combined dataset of 933 images from the (Tushabe et al., 2024) Ugandan clinical microscopy dataset and the BCCD dataset, ResNet-50 with unfrozen transfer learning achieved 98.57% accuracy, AUC-ROC of 0.9973, sensitivity of 98.41%, and specificity of 98.70%, with only 2 errors on 140 test images. Grad-CAM explainability analysis confirmed that the model attends to biologically meaningful regions, and both errors were attributable to specific and identifiable causes.

The results demonstrate that transfer learning from ImageNet effectively bridges the data scarcity challenge inherent in medical imaging research in low-resource settings. The performance gap between the baseline CNN at 87.86% accuracy and ResNet-50 at 98.57% accuracy quantifies this benefit for the blood smear microscopy domain.

This study makes a specific and verifiable contribution: to the best of our knowledge, it is the first deep learning study applied to the (Tushabe et al., 2024) Ugandan clinical mobile-phone microscopy dataset. The deployed Gradio application at <https://huggingface.co/spaces/Patrovas/sickle-cell-detection> provides a working proof of concept for a mobile-first, non-specialist sickle cell screening tool designed for the realities of Ugandan rural healthcare.

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